Arista Access Node KR-ACC-24 KR Puram Access Hardware/Environmental Fault Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2854

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults on Arista Access Node KR-ACC-24 located at KR Puram Access. The failure domain in question, Hardware/Environmental Fault, is primarily caused by ASIC memory parity errors or line card crashes due to power supply redundancy loss. This specific failure domain occurs on Arista Access Nodes because of their reliance on ASICs (Application-Specific Integrated Circuits) to process and forward traffic. In the event of an ASIC memory parity error, the node will fail to forward traffic, resulting in a minor alarm (ALARM_2854). Additionally, power supply redundancy loss can cause line card crashes, further exacerbating the issue. The root cause of this failure domain is often attributed to environmental factors such as overheating, power supply issues, or hardware defects. This SOP outlines the steps to diagnose and resolve these issues, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Verify remote hands are dispatched to KR Puram Access to assist with the resolution process.
- Check redundant link status to ensure no impact on network connectivity.
- Gather logs and configuration files for analysis.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash due to power supply redundancy loss	Medium
Hardware defects or environmental factors	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal asic detail KR-ACC-24, Expected Output: ASIC memory parity error detected, Description: This command isolates the ASIC memory parity error, which is a primary root cause of this failure domain.

Command: show system internal power detail KR-ACC-24, Expected Output: Power supply redundancy loss detected, Description: This command identifies power supply redundancy loss, which can cause line card crashes and exacerbate the issue.

Command: show system internal log KR-ACC-24, Expected Output: Log entries indicating hardware defects or environmental factors,
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Description: This command gathers log entries that may indicate hardware defects or environmental factors contributing to the failure domain.
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Resolution Steps

Step 1: Step: Replace faulty ASIC, Command: hardware ASIC KR-ACC-24 replace, Description: Replace the faulty ASIC to resolve the memory parity error and restore normal operation.

Step 2: Step: Replace power supply unit, Command: hardware power KR-ACC-24 replace, Description: Replace the power supply unit to resolve power supply redundancy loss and prevent line card crashes.

Step 3: Step: Perform environmental checks and maintenance, Command: environmental KR-ACC-24 check, Description: Perform environmental checks and maintenance to ensure optimal operating conditions and prevent future failures.

Post-Check Verification

```
Command: show system internal asic detail KR-ACC-24, Expected Output: No ASIC memory parity error detected, Description: Verify that the ASIC memory parity error has been resolved.

Command: show system internal power detail KR-ACC-24, Expected Output: Power supply redundancy intact, Description: Verify that the power supply redundancy has been restored.

Command: show system internal log KR-ACC-24, Expected Output: No log entries indicating hardware defects or environmental factors, Description: Verify that no hardware defects or environmental factors are contributing to the failure domain.
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Roll-back Procedures

- Step: Restore original configuration, Command: configure replace KR-ACC-24, Description: Restore the original configuration to ensure that all changes made during the resolution process are reversed.

- Step: Reboot node, Command: system reboot KR-ACC-24, Description: Reboot the node to ensure that all changes made during the resolution process are propagated.